```
  GNU nano 7.2                                                                                   p3t1.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    printf("===== Process Before fork() =====\n");
    printf("PID  : %d\n", getpid());
    printf("PPID : %d\n", getppid());

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed!\n");
        exit(1);
    }

    else if (pid == 0)
    {
        // Child Process
        printf("\n===== Child Process =====\n");
        printf("Child PID  : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        printf("Child is running...\n");
        sleep(5);

        printf("Child process terminating...\n");
        exit(0);
    }

    else
    {
        // Parent Process
        printf("\n===== Parent Process =====\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID  : %d\n", pid);

        printf("Parent is waiting for child...\n");

        wait(NULL);

        printf("Child process completed.\n");
        printf("Parent process terminating...\n");
    }

    return 0;
```

```
mohana@MOHANA:~$ nano p3t1.c
mohana@MOHANA:~$ gcc p3t1.c -o p3t1
mohana@MOHANA:~$ ./p3t1
===== Process Before fork() =====
PID  : 1523
PPID : 320

===== Parent Process =====
Parent PID : 1523
Child PID  : 1524
Parent is waiting for child...

===== Child Process =====
Child PID  : 1524
Parent PID : 1523
Child is running...
Child process terminating...
Child process completed.
Parent process terminating...
mohana@MOHANA:~$
```

TASK-2

```
 GNU nano 7.2                                                         p3t2.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed!\n");
        exit(1);
    }

    else if (pid == 0)
    {
        // Child Process
        printf("Child Process Started\n");
        printf("Child PID  : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        printf("Child is running...\n");

        // Child enters sleeping/waiting state
        printf("Child is going to sleep...\n");
        sleep(20);

        printf("Child woke up and is running again...\n");

        sleep(5);

        printf("Child terminating...\n");
        exit(0);
    }

    else
    {
        // Parent Process
        printf("Parent Process Started\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID  : %d\n", pid);

        printf("Parent is waiting for child...\n");

        // Parent waits for child
        wait(NULL);

        printf("Child terminated.\n");
        printf("Parent terminating...\n");
    }

    return 0;
}
```

```
mohana@MOHANA:~$ nano p3t2.c
mohana@MOHANA:~$ gcc p3t2.c -o p3t2
mohana@MOHANA:~$ ./p3t2
Parent Process Started
Parent PID : 1566
Child PID  : 1567
Parent is waiting for child...
Child Process Started
Child PID  : 1567
Parent PID : 1566
Child is running...
Child is going to sleep...
Child woke up and is running again...
Child terminating...
Child terminated.
Parent terminating...
mohana@MOHANA:~$ ps -ef | grep task
mohana      1571     320  0 13:27 pts/0    00:00:00 grep --color=auto task2
mohana@MOHANA:~$ ps -o pid,ppid,state,stat,cmd -p <PID>
-bash: syntax error near unexpected token `newline'
mohana@MOHANA:~$ ps -o pid,ppid,state,stat,cmd -p 2451
    PID    PPID S STAT CMD
mohana@MOHANA:~$ cat /proc/2451/status
cat: /proc/2451/status: No such file or directory
mohana@MOHANA:~$
```